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Droplet spreading on a rotating disc
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Introduction

Incompressible Navier-Stokes Equations

Momentum Equation:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \mathbf{f} \tag{1}$$

Incompressibility Condition:

$$\nabla\cdot\mathbf{u} = 0 \tag{2}$$

Where:

- $\mathbf{u}$ = velocity field (vector)
- p = pressure scalar
- ρ = density (constant for incompressible fluid)
- μ = dynamic viscosity
- $\mathbf{f}$ = body force density (e.g., gravity)

For a small viscous drop with

$$\mathrm{Oh} \gg 1,\ \mathrm{Bo} \ll 1, \text{ and } \mathrm{Re} \ll 1$$

the evolution of the liquid film is governed primarily by viscosity, capillarity, and centrifugal force, while the effects of inertia, gravity, and Coriolis forces can be neglected. Under these assumptions in small aspect ratio, the governing Navier–Stokes equations systematically reduce to the axisymmetric thin film equation for droplet spreading on a rotating disc.

Radial Thin Film Equation

This section introduces the axisymmetric thin-film equation governing the spreading of a viscous droplet on a rotating disc. We consider the axisymmetric thin film equation from Li Wang paper [2], spin coating dynamics are highly dependent on rotational speed. Thin-film equations have been analyzed extensively in the context of capillarity-driven flows and lubrication asymptotics. In the present work, we adopt a radial formulation that captures the coupling between curvature-driven pressure and mobility, and incorporate a rotational Bond number to account for centrifugal effects. The resulting PDE, stated below, serves as the starting point for our analysis and discretization.

$$\boxed{u_t + \frac{1}{r}\left\{ru^2(u+3\lambda)\left[\left(\frac{(ru_r)_r}{r}\right)_r + \beta r\right]\right\}_r = 0,\ \Omega\subset\mathbb{R},\ t\in(0,T)} \tag{3}$$

Where:

$$\beta = \frac{\rho\omega^2 V}{2\pi\sigma\theta_e^2} \quad \text{(rotational Bond number)}$$

- u = Droplet height
- r = Radius covered by the Droplet
- R = Radius of the substrate/Disc
- ω = Angular velocity
- λ = Slip parameter

Simplified Equation

We start from the full thin-film equation and simplify it further by removing the centrifugal term (i.e., βr), focusing solely on capillary-driven spreading. The problem is then posed with an initial condition and symmetry-preserving boundary conditions at the center and no-flux boundaries at r = R. This reduced PDE highlights the nonlinear diffusion mechanism where the mobility M(u) depends on the film thickness:

$$\boxed{u_t + \frac{1}{r}\left\{ru^2(u+3\lambda)\left(\frac{(ru_r)_r}{r}\right)_r\right\}_r = 0} \tag{4}$$

Initial Condition:

$$u(r,0) = u_0(r) \quad \text{in } \Omega$$

Boundary Conditions:

$$\text{At } r = 0 \text{ (center)}: \quad u_r = 0, \quad u_{rrr} = 0$$
$$\text{At } r = R \text{ (Outer Boundary)}: \quad u_r = 0, \quad u_{rrr} = 0$$

See Appendix 1 for more details.

General Form from Literature

From Grün and Rumpf , a general form of the thin-film equation in higher dimensions is presented. This compact representation captures capillarity-driven evolution. Reduction to the 1D (axisymmetric) case yields a simpler structure with mobility $M(u) = u^3$ for a no-slip surface. This connection shows that our radial system is a special case of a more general fourth-order PDE.

From the Grün and Rumpf paper [1], the thin film equation in higher dimensions is:

$$u_t + \operatorname{div}(M(u)\nabla\Delta u) = 0 \quad \text{on } \Omega \subset \mathbb{R},\ t \in (0,T) \tag{5}$$

Boundary conditions:

$$\frac{\partial u}{\partial \nu} = \frac{\partial \Delta u}{\partial \nu} = 0 \quad \text{on } \partial\Omega$$
$$u(\nu,0) = u_0(\nu) \quad \text{in } \Omega$$

Reduction to 1D (Axisymmetric)

Reducing equation (5) to the 1D axisymmetric case, we get:

$$\boxed{u_t + \frac{\partial}{\partial x}\left(M(h)\frac{\partial^3 u}{\partial x^3}\right) = 0 \quad x \in \Omega,\ t \in (0,T)} \tag{6}$$

with boundary condition (considered no-slip):

$$\frac{\partial u}{\partial r} = 0 = \frac{\partial^3 u}{\partial r^3} \quad \text{on } \partial\Omega$$

Here $M(u)$ is the mobility; under no-slip assumption $M(u) = u^3$.

Rewriting Equation (4) in Terms of P

To bring the equation into a form suitable for finite element discretization, the pressure-like term P is introduced, defined as a scaled radial Laplacian of u. The reformulation:

$$\boxed{u_t + \frac{1}{r}\left(rM(u)P_r\right)_r = 0, \quad \text{where } P = \frac{(ru_r)_r}{r}, \quad M(u) = u^2(u+3\lambda)} \tag{7}$$

makes the structure more symmetric and allows the two unknowns u and P to be treated via coupled weak formulations, which is advantageous for stability and energy estimates.

Variational Form of Equation (7)

The PDE is transformed into variational (weak) form by multiplication with an arbitrary test function and integration by parts, applying the no-flux boundary conditions. This leads to two weak formulations: one for the main equation involving u and P_r, and another for P itself in terms of u_r.

Transforming equation (7) into variational form with arbitrary test function θ, we have:

$$\int_0^R \left[u_t + \frac{1}{r}(rM(u)P_r)_r\right] \theta\, r\, dr = 0$$

$$\int_0^R u_t \theta r\, dr - \int_0^R rM(u)P_r\theta_r\, dr = 0$$

$$\int_0^R (rM(u)P_r)_r\, \theta\, dr = \Big[rM(u)P_r\, \theta\Big]_0^R - \int_0^R rM(u)P_r\, \theta_r\, dr$$

(integration by parts + applying b.c.s)

$$\int_0^R u_t\, \theta\, r\, dr - \int_0^R rM(u)P_r\, \theta_r\, dr = -\Big[rM(u)P_r\, \theta\Big]_0^R$$

$$\boxed{\int_0^R u_t r\theta\, dr = \int_0^R rM(u)P_r\theta_r\, dr \quad \forall\theta \in V := H_0^1(\Omega)} \tag{8}$$

Considering the weak form for P, we have:

$$\int_0^R P\psi r\, dr = \int_0^R \left(\frac{(ru_r)_r}{r}\right) \psi r\, dr \quad \forall\psi \in V$$

(integration by parts + applying b.c.s)

$$\int_0^R (ru_r)_r\, \psi\, dr = [ru_r\psi]_0^R - \int_0^R ru_r\, \psi_r\, dr$$

$$\boxed{\int_0^R P\psi r\, dr = -\int_0^R ru_r\psi_r\, dr \quad \forall\psi \in V} \tag{9}$$

These weak forms are the basis for Galerkin finite element discretization and later stability of solution.

Discretization

Let h be the spatial discretization parameter, and $V^h \subset V$ on $[0, R]$. Let $\{\phi_j(r)\}$ be the basis in V^h.

$$U^{n+1}(r) = \sum_j U_j^{n+1}\phi_j(r), \qquad P^{n+1}(r) = \sum_j P_j^{n+1}\phi_j(r)$$

Discrete inner products are defined with radial weighting, and separable mass and stiffness matrices are introduced to handle the geometry-dependent terms.
Defining the discrete inner product:

$$(u, v)_h = \int_0^R ruv\, dr = \sum_{ij} ru_iv_j \int_0^R \phi_i(r)\phi_j(r)\, dr$$

Let τ be the time step. Using implicit Euler:

$$u^t = \frac{U^{n+1} - U^n}{\tau}$$

Discrete Weak Form

$$\left(\frac{U^{n+1}-U^n}{\tau}, r\theta\right)_h - \int_0^R rM(U^{n+1})P_r^{n+1}\theta_r\,dr = 0 \quad \forall\theta \in V^h \tag{10}$$

$$(P^{n+1}, r\psi)_h = -\int_0^R rU_r^{n+1}\psi_r\,dr \quad \forall\psi \in V^h \tag{11}$$

This results in a coupled nonlinear system for U^{n+1} and $P_n + 1$ at each time step.

Discrete Energy Estimate

This section derives the discrete counterpart of the continuous energy dissipation law. In the continuous model, the L^2-norm of the gradient of u decreases over time, with energy dissipated through the nonlinear mobility term. The discrete derivation follows by taking -P^{n+1} as the test function, manipulating sums via the weak form of P and summing over time steps.

Energy estimate in continuous setting:

$$\frac{1}{2}\int_0^R (u_r(T,r))^2\,r\,dr + \int_0^T\int_0^R M(u)(P_r)^2\,r\,dr\,dt = \frac{1}{2}\int_0^R (u_{0,r}(r))^2\,r\,dr$$

Discrete setting:

For the discrete weak form we consider the test function θ as $-P^{n+1}$

$$\left(\frac{U^{n+1}-U^n}{\tau},\ -rP^{n+1}\right)_h - \int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)\left(-P_r^{n+1}\right)dr = 0$$

$$-\left(\frac{U^{n+1}-U^n}{\tau},\ rP^{n+1}\right)_h + \int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)^2 dr = 0$$

$$\left(\frac{U^{n+1}-U^n}{\tau},\ rP^{n+1}\right)_h = \int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)^2 dr$$

Using Weak Form of P

$$(rP^{n+1},\, U^{n+1}-U^n)_h = -\int_0^R rU_r^{n+1}(U_r^{n+1}-U_r^n)dr \tag{12}$$

Expand the right side using $a(a-b) = \frac{1}{2}[a^2 - b^2 + (a-b)^2]$:

$$(rP^{n+1},\, U^{n+1}-U^n)_h = -\frac{1}{2}\int_0^R r\left[(U_r^{n+1})^2 - (U_r^n)^2 + (U_r^{n+1}-U_r^n)^2\right]dr \tag{13}$$

Therefore,

$$\int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)^2 dr = -\frac{1}{\tau}\frac{1}{2}\int_0^R r\left[(U_r^{n+1})^2 - (U_r^n)^2 + (U_r^{n+1}-U_r^n)^2\right]dr \tag{14}$$

Multiply both sides by τ and sum from $n = 0$ to $N-1$:

$$\sum_{n=0}^{N-1}\tau\int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)^2 dr = -\frac{1}{2}\int_0^R r(U_r^N)^2dr + \frac{1}{2}\int_0^R r(U_r^0)^2dr$$
$$-\frac{1}{2}\sum_{n=0}^{N-1}\int_0^R r(U_r^{n+1}-U_r^n)^2dr \tag{15}$$

Rearranged, the fully **discrete energy estimate** is:

$$\boxed{\frac{1}{2}\int_0^R r(U_r^N)^2 dr + \frac{1}{2}\sum_{n=0}^{N-1}\int_0^R r(U_r^{n+1}-U_r^n)^2 dr + \sum_{n=0}^{N-1}\tau\int_0^R rM(U^{n+1})\left(P_r^{n+1}\right)^2 dr = \frac{1}{2}\int_0^R r(U_r^0)^2 dr} \tag{16}$$

The final estimate confirms that the numerical scheme is energy stable and that energy decreases monotonically in time.

Discrete Entropy Estimate

In addition to energy, the thin-film model has an entropy functional related to the degeneracy of the mobility $M(u)$ at u=0. This section introduces entropy functional and shows how to obtain a discrete entropy inequality, mimicking the continuous setting.

Entropy/pressure estimate in continuous setting:

$$\int_0^R G(u(T,r))\, r\, dr + \int_0^T\int_0^R P^2\, r\, dr\, dt = \int_0^R G(u_0(r))\, r\, dr$$

$$G(u) = \int_A^u \left(\int_A^s \frac{1}{M(b)}\, db\right) ds, \quad \text{where } M(b) = b^2(b+3\lambda).$$

Discrete setting:
Define the entropy functional:

$$G(u) := \int_1^u \int_1^s \frac{1}{M(b)}\, db\, ds, \quad \text{with } M(b) = b^2(b+3\lambda)$$

Now choose test function:

$$\theta = I_h G'(U^{n+1})$$

$$\left(\frac{U^{n+1}-U^n}{\tau}, I_h G'(U^{n+1})r\right)_h - \int_0^R rM(U^{n+1})\nabla P^{n+1}\cdot\nabla(I_h G'(U^{n+1}))\, dr = 0 \tag{17}$$

From the structure of $G(u)$:

$$G'(u) = \int_0^u \frac{1}{M(\xi)}\, d\xi \quad \Rightarrow \quad G''(u) = \frac{1}{M(u)}$$

So:

$$rM(U^{n+1})\nabla\left(I_h G'(U^{n+1})\right) = r\nabla U^{n+1}$$

$$\Rightarrow \int_0^R r\nabla P^{n+1}\cdot\nabla U^{n+1}\, dr = \int_0^R rP^{n+1}P_r^{n+1}\, dr \quad \text{(using equation (11))}$$

Then:

$$\boxed{\left(rI_h\left(G(U^N)-G(U^0)\right),1\right)_h + \left(rP^{n+1},P_r^{n+1}\right)_h \le 0} \tag{14}$$

Summing up from $n=0$ to $n-1$, multiplying by τ and using the fact that $(I_h\eta,1)_h = \int_\Omega I_h\eta\, dr$, we obtain the results. This entropy estimate is crucial for proving nonnegativity and for controlling the solution near regions where the film becomes very thin.

Nonnegativity results for discrete solutions

In this section, we aim to establish the *existence of nonnegative discrete solutions* to our radial thin-film model by leveraging the discrete entropy estimate derived earlier (Equation 14). Following the framework introduced by Grün and Rumpf [1], we consider modified discrete mobilities M_σ, where $\sigma > 0$ is a small control parameter. This leads to a corresponding discrete entropy functional G_σ that satisfies the key inequality:

$$\left(rI_h\left(G(U_\sigma^N) - G(U_\sigma^0)\right), 1\right)_h + \left(rP_\sigma^{n+1}, P_\sigma^{n+1}\right)_h \leq 0$$

This estimate ensures that the *discrete entropy does not increase* over time, and allows us to show that for sufficiently small σ, the solution remains bounded from below:

$$U_{\tau h}^\sigma \geq -\varepsilon \quad \text{on } \Omega_T$$

for arbitrary $\varepsilon > 0$, i.e., *nonnegative up to controllable error*. Moreover, if the mobility $M(u) = u^2(u + 3\lambda)$ satisfies a growth condition with exponent $n > 2$, we can further prove that solutions remain *strictly positive* if the initial condition is strictly positive. This approach mirrors the continuous results from entropy methods and confirms that our scheme aligns with the nonnegativity-preserving framework proposed in [1].

Existence of Nonnegative Discrete Solutions

Let $\mathcal{T}_h$ be an admissible discretization of the radial domain $\Omega = [0, R]$ used in our formulation of the thin-film equation, and let $n = 3$ be the growth exponent of the mobility function $M(u) = u^2(u + 3\lambda)$. Note that this mobility is nonnegative and vanishes at $u = 0$, satisfying the assumptions from [1].

For any given $\varepsilon > 0$, there exists a positive control parameter $\sigma_0 > 0$, depending on d, n, ε, h, and the initial data $u_0 \geq 0$, such that:

For every $0 < \sigma < \sigma_0$, discrete entropy–mobility pairs (G_σ, M_σ) can be constructed with the property that the discrete solutions $U_{\tau h}^\sigma$ of our system (see Equation 14) satisfy:

- $U_{\tau h}^\sigma > \sigma/2 \quad \text{if} \quad u_0 \geq \sigma_0 \quad \text{and} \quad n > 2\,(n = 3 \text{ in our case})\,.$

This result confirms that our discrete scheme preserves nonnegativity and even strict positivity under appropriate initial conditions and sufficiently small control parameter σ. The theorem is adapted from Theorem 6.1 in [1], where a general theory is developed for constructing nonnegativity-preserving finite element schemes for fourth-order degenerate parabolic equations.

Nonnegativity and Positivity Preservation for $n = 3$

Our mobility function $M(u) = u^2(u + 3\lambda)$ grows like u^3, so the exponent is $n = 3 > 2$. Following the theory developed by Grün and Rumpf [1], we are in the case where strict positivity of the discrete solution can be established, provided the initial data is strictly positive.

To control the behavior of the discrete solution, we consider the cardinality of the set of "bad" nodes where the solution could dip below a threshold $\beta < 0$:

$$K_\beta(t) := \{x_i \in \Omega : U_{\tau h}^\sigma(t, x_i) < \beta\}$$

where, x_i is the nodal point.
The paper [1] provides the following upper bound for $d = 1$:

$$|K_\beta(t)| \leq \frac{c_1^{-1} C_{\text{entrop}}\, \sigma}{h\, R_2^M(\beta)}$$

where $R_2^M(\beta)$ is the second primitive of $1/M(u)$. Since $R_2^M(\beta) \to \infty$ as $\beta \to 0^-$, we conclude that $|K_\beta(t)| \to 0$ as $\sigma \to 0$. This means for small enough σ, the discrete solution becomes strictly positive at all nodes and times:

$$U_{\tau h}^\sigma(t, x_i) > 0 \quad \text{for all } x_i \in \Omega,\ t \in [0, T]$$

To implement this numerically, the discrete mobility $M_\sigma(U)|_E$ is defined piecewise depending on the signs of values at adjacent nodes U_1, U_2. According to Equation (30) in [1], the definition is:

$$M_\sigma(U)\big|_E := \begin{cases} \dfrac{(1-n)(U_2-U_1)}{U_2^{1-n}-U_1^{1-n}} & \text{if } \min(U_1,U_2) \geq 0 \\ \dfrac{(1-n)(U_2-U_1)}{\sigma^{-1}|U_1|^{1-n}+U_2^{1-n}} & \text{if } U_1 < 0 < U_2 \\ \dfrac{\sigma(n-1)(U_2-U_1)}{|U_1|^{1-n}-|U_2|^{1-n}} & \text{if } \max(U_1,U_2) \leq 0 \\ M(U_1) & \text{if } U_1 = U_2 \geq 0 \\ \sigma M(-U_1) & \text{if } U_1 = U_2 < 0 \end{cases}$$

In our radial discretization of the thin-film equation, the 1D setting is sufficient. The generalization to higher dimensions (as shown in the subsequent section of [1]) defines element-wise mobility $M(U)|_E$ via entropy derivatives and orthogonal transformations. However, this construction is not required in our case and can be omitted.

We conclude that our discrete scheme, equipped with the entropy estimate and mobility regularization as described above, ensures the existence of strictly positive discrete solutions for all time steps.

Centrifugal term

To incorporate rotation, we apply an operator-splitting step each time step t. First, starting from u_k, we advance only the capillarity–mobility part (set = 0) to obtain an intermediate state $u_{k+1/2}$; here the mobility M(u) is frozen at u_k and we solve the resulting linear implicit system (with the usual symmetry/no-flux boundary conditions), which captures fourth-order smoothing. Next, using $u_{k+1/2}$ as input, we evolve the centrifugal transport subproblem over the same t with an Engquist–Osher upwind flux, choosing the upwind direction by the sign of f(u,r).

Considering 'β' or centrifugal part & using splitting technique:

$$v_t + \frac{1}{r}\frac{\partial}{\partial r}\left[ru^2(u+3\lambda)\beta r\right] = 0 \tag{18}$$

Define the flux function:

$$f(u,r) = \beta r^2 u^2 (u+3\lambda)$$

Compute its derivative with respect to u:

$$f'(u,r) = 3\beta r^2 u^2 + 6\beta r^2 u\lambda$$

Using Engquist-Osher Scheme(Upwinding):

$$u_i^{k+1} = u_i^k - \frac{\Delta t}{\Delta x}\begin{cases} f(u_i^k) - f(u_{i-1}^k) & \text{if } f'(u,r) > 0 \\ f(u_{i+1}^k) - f(u_i^k) & \text{if } f'(u,r) \leq 0 \end{cases} \tag{19}$$

This two-stage update decouples stiff diffusion from advection: the first stage preserves the energy/entropy structure via a stable linear solve, while the second stage updates the rotational flux explicitly.

Algebraic Form

Considering eq. (8) and for simplicity making it linear and substituting the basis function estimates into the unknown

we get:

$$\sum_{i=1}^{N}\left(\sum_{j=1}^{N}\frac{U_j^{n+1}-U_j^n}{\tau}(\varphi_j, r\varphi_i)_h - \sum_{j=1}^{N}P_j^{n+1}\int_0^R rM(U^n(r))\varphi_j'(r)\varphi_i'(r)dr\right) = 0 \tag{20}$$

$$\Rightarrow \quad B_h \frac{U_j^{n+1} - U_j^n}{\tau} - L_h(U^n)P^{n+1} = 0$$

Similarly for eq. (9):

$$\sum_{j=1}^{N} P_j^{n+1}(\varphi_j, r\varphi_i)_h = -\sum_{j=1}^{N} U_j^{n+1} \int_0^R r\varphi_j'(r)\varphi_i'(r)dr$$

$$\Rightarrow \quad B_h P^{n+1} = -L_h U^{n+1}$$

$$\Rightarrow \quad \begin{pmatrix} \mathbf{B}_h & -\tau \mathbf{L}_h(U^n) \\ \mathbf{L}_h & \mathbf{B}_h \end{pmatrix} \begin{pmatrix} \mathbf{U}^{n+1} \\ \mathbf{P}^{n+1} \end{pmatrix} = \begin{pmatrix} \mathbf{B}_h \mathbf{U}^n \\ 0 \end{pmatrix} \tag{21}$$

where:

$$\text{lumped mass matrix } (B_h)_{ij} = (\varphi_j, r\varphi_i)_h = \int_0^R r\varphi_j\varphi_i$$

$$\text{stiffness matrix } (L_h)_{ij} = (\varphi_j', r\varphi_i')_h = \int_0^R r\varphi_j'(r)\varphi_i'(r)dr$$

$$\text{stiffness matrix (Mobility weighted) } (L_h(U))_{ij} = \int_0^R rM(U^n(r))\varphi_j'(r)\varphi_i'(r)dr$$

Since we considered piecewise linear shape function:

$$\varphi_j(r) = \begin{cases} \frac{r - r_{j-1}}{h} & \text{for } r \in [r_{j-1}, r_j] \\ \frac{r_{j+1} - r}{h} & \text{for } r \in [r_j, r_{j+1}] \\ 0 & \text{elsewhere} \end{cases} \tag{22}$$

$$\varphi_j'(r) = \begin{cases} \frac{1}{h} & \text{for } r \in [r_{j-1}, r_j] \\ -\frac{1}{h} & \text{for } r \in [r_j, r_{j+1}] \\ 0 & \text{elsewhere} \end{cases} \tag{23}$$

Thus our matrices can be approximated as:

- **lumped mass matrix approximation:** $(B_h)_{ij} \approx \int_{r_i} r\varphi_j(r)dr \approx r_i \cdot h$ (midpoint rule)
- **Stiffness matrix approximation:** $(L_h)_{ij} \approx -\frac{r_j + r_i}{2h}$
- **Mobility weighted Stiffness:** $(L_h(U))_{ij} \approx M \cdot (L_h)_{ij}$

This block system is linear in $\mathbf{U}^{n+1}$ and $\mathbf{P}^{n+1}$ at each time step, with the coefficient matrix depending only on known quantities from $\mathbf{U}^n$. It can be efficiently solved using linear algebra libraries, such as numpy.linalg.solve in Python, by assembling the matrices and solving the combined vector equation.

Numerical Results

In this section, we present numerical simulations of the radial thin-film equation derived in the previous sections. The results are compared under different slip conditions, entropy–mobility frameworks, and centrifugal effects, following the nonnegativity-preserving framework of Grün and Rumpf [1].

Case 1: No Slip ($\lambda = 0$), No Entropy–Mobility Pair

Without the entropy–mobility pair, the numerical solution violates nonnegativity. The droplet height profile dips below zero, indicating instability of the scheme in degenerate regions. This demonstrates the need for entropy stabilization to preserve the physical constraint $u \geq 0$.

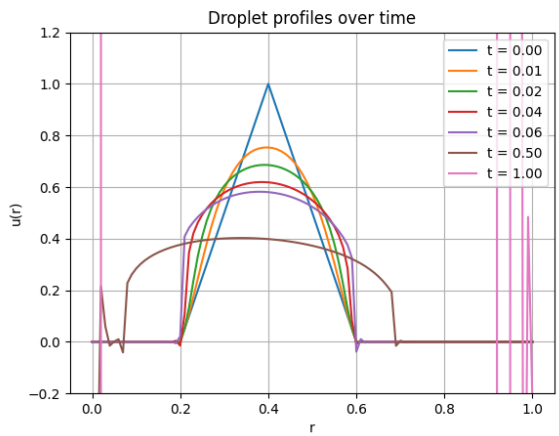

Figure 1: Droplet spreading without slip ($\lambda = 0$) and without entropy–mobility stabilization. Negative values occur in thin regions.

Case 2: No Slip ($\lambda = 0$), With Entropy–Mobility Pair

When the entropy–mobility pair is incorporated, the discrete entropy estimate prevents unphysical negative values of the droplet height. The solution remains nonnegative throughout the simulation, confirming the stabilizing role of the entropy functional as described by Grün and Rumpf [1].

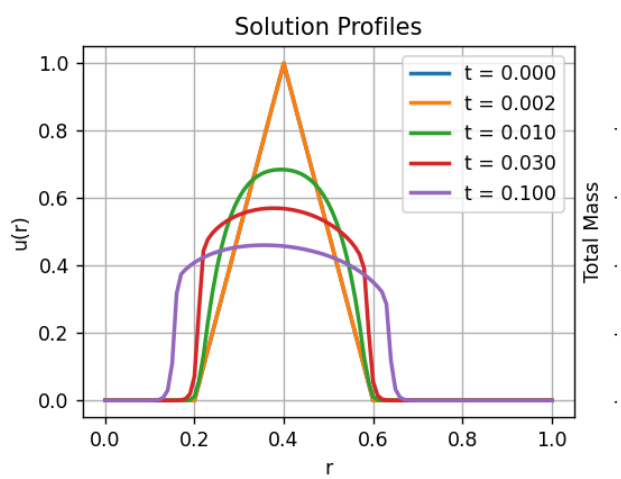

Figure 2: Droplet spreading without slip ($\lambda = 0$) but with entropy–mobility stabilization. Nonnegativity is preserved.

Case 3: Slip Condition ($\lambda = 4$), With Entropy–Mobility Pair

With the slip condition $\lambda = 4$, the spreading is accelerated compared to the no-slip case. However, thanks to the entropy–mobility stabilization, the solution remains nonnegative and stable throughout the simulation. This confirms that the entropy framework guarantees positivity even when slip effects are included.

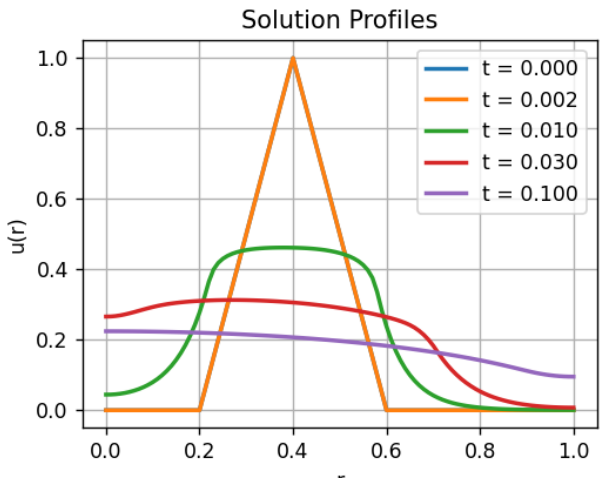

Figure 3: Droplet spreading with slip condition ($\lambda = 4$) and entropy–mobility stabilization. The scheme preserves nonnegativity under slip effects.

Case 4a: Centrifugal Term ($\beta = 1$)

For a moderate centrifugal effect ($\beta = 1$), the droplet spreads outward more strongly compared to the purely capillarity-driven case. The film height profile remains nonnegative due to the entropy–mobility stabilization, while the radial spreading is enhanced by the rotational force.

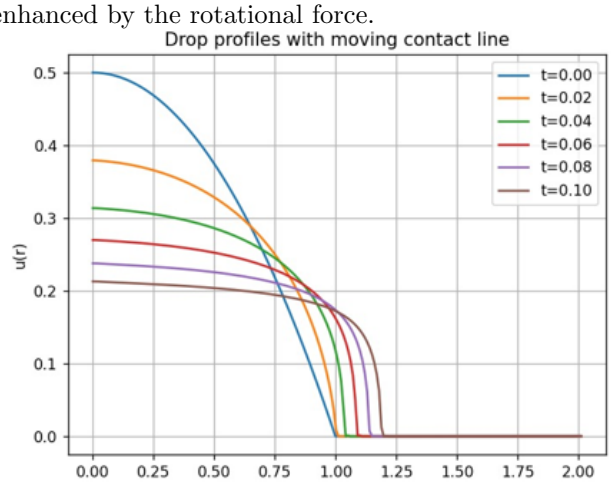

Figure 4: Droplet spreading with centrifugal term $\beta = 1$. Outward spreading is enhanced while nonnegativity is preserved.

Case 4b: Centrifugal Term ($\beta = 2$)

Increasing β to 2 strengthens the centrifugal effect. The droplet spreads faster outward, leading to a thinner film at the center. The entropy–mobility pair ensures stability and prevents rupture in the thin regions.

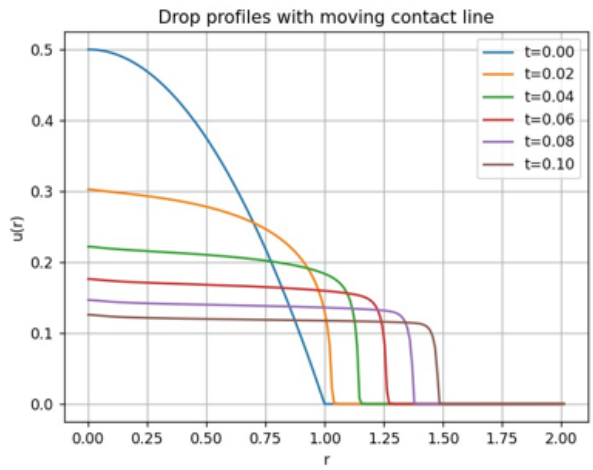

Figure 5: Droplet spreading with centrifugal term $\beta = 2$. Stronger rotation accelerates spreading while stability is preserved.

Case 4c: Centrifugal Term ($\beta = 4$)

For a strong centrifugal force ($\beta = 4$), the droplet rapidly spreads outward from the center. The film thickness at the center decreases quickly and the profile shows significant thinning over time. Small oscillations appear at the advancing front due to the strong centrifugal force, but the entropy–mobility scheme maintains overall stability and prevents rupture.

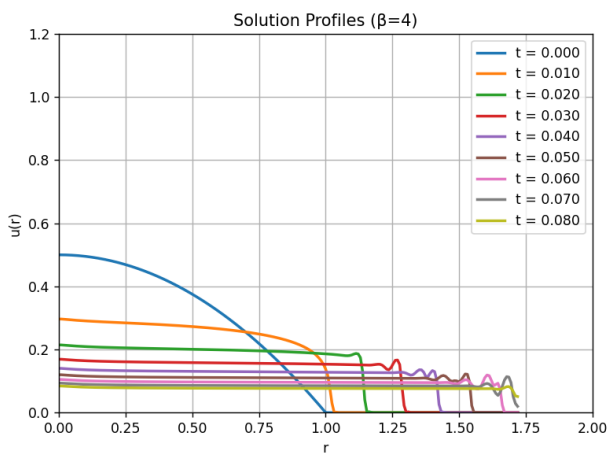

Figure 6: Droplet spreading with centrifugal term $\beta = 4$.

Entropy Estimate

The entropy functional $G(u)$ introduced by Grün and Rumpf [1] is expected to decrease monotonically over time, providing a discrete control mechanism that ensures the non-negativity of the numerical solution.

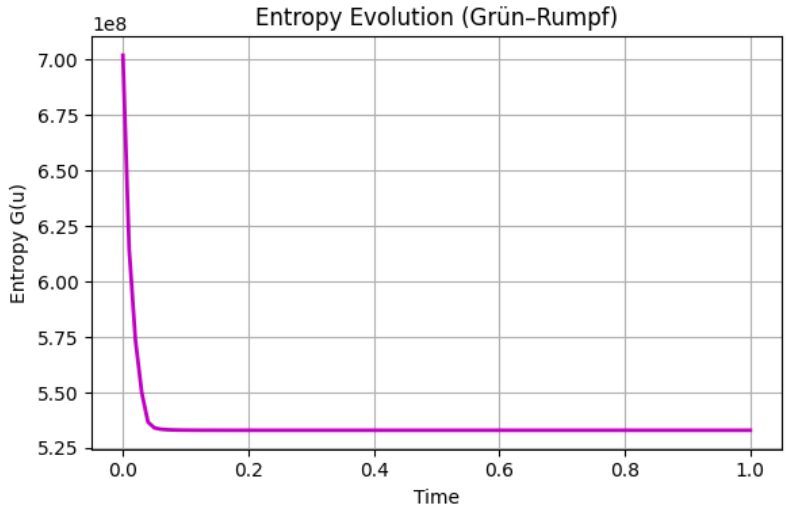

Figure 7: Entropy evolution during droplet spreading.

As seen in Fig. 7, the entropy decreases steeply in the initial phase and then remains constant. This behavior exactly matches the theoretical framework of Grün–Rumpf, confirming that the implemented scheme preserves nonnegativity.

Conclusion

In this seminar report, we studied the spreading of a viscous droplet on a rotating disc, starting from the incompressible Navier–Stokes equations and systematically reducing them to the axisymmetric thin-film equation. A variational formulation and its finite element discretization were derived, and discrete energy and entropy estimates were established to guarantee stability and nonnegativity of the numerical solutions.

The numerical experiments demonstrated that without entropy–mobility stabilization, the solution develops nonphysical negative values, whereas the entropy–mobility framework ensures stability and preserves nonnegativity under both no-slip and slip conditions. Furthermore, the inclusion of centrifugal effects via an operator-splitting scheme captured the enhanced outward spreading at increasing rotational Bond numbers.

Overall, the analysis and simulations confirm that the entropy–mobility approach of Grün and Rumpf provides a robust framework for constructing stable and physically consistent numerical schemes for degenerate thin-film equations with rotational effects. Future work may extend this framework to higher dimensions and explore adaptive discretizations for improved efficiency.

Appendix

Boundary Conditions from Mass Conservation

We consider the axisymmetric thin-film equation without external forces:

$$u_t + \frac{1}{r}\left(rM(u)P_r\right)_r = 0, \quad \text{where } P = \frac{(ru_r)_r}{r}, \quad M(u) = u^2(u+3\lambda)$$

To ensure global conservation of mass, we require:

$$\frac{d}{dt}\int_0^R u(r,t)\, r\, dr = 0.$$

Using the PDE, we substitute u_t and integrate by parts:

$$\frac{d}{dt}\int_0^R u\, r\, dr = -\int_0^R \left\{\frac{1}{r}\frac{d}{dr}\left[rM(u)\cdot p_r\right]\right\} r\, dr = -\left[rM(u)\, p_r\right]_{r=0}^{r=R},$$

where we define the radial Laplacian as $p := \Delta_r u = \frac{1}{r}\frac{d}{dr}(ru_r)$.
Thus, conservation of mass requires that:

$$\left[rM(u)\, p_r\right]_{r=0}^{r=R} = 0.$$

At the outer boundary $r = R$, one may impose either $u(R) = 0$ or $p_r(R) = 0$ to make the flux vanish.
To justify the vanishing of the flux at the center $r = 0$, we evaluate the limit:

$$\lim_{r\to 0} rM(u(r))p_r(r).$$

Assuming the solution $u(r)$ is smooth and symmetric, we expand it using Taylor series:

$$u(r) = u(0) + \frac{1}{2}u_{rr}(0)\, r^2 + \frac{1}{6}u_{rrr}(0)\, r^3 + \mathcal{O}(r^4),$$

which implies:

$$u_r(r) = u_{rr}(0)\, r + \frac{1}{2}u_{rrr}(0)\, r^2 + \mathcal{O}(r^3), \quad p_r(r) = \mathcal{O}(r^1).$$

Thus,

$$rM(u(r))p_r(r) = \mathcal{O}(r),$$

and hence:

$$\lim_{r\to 0} rM(u)p_r = 0.$$

This shows that the flux vanishes at $r = 0$ provided the solution is smooth. To guarantee this, we impose the symmetry boundary conditions:

$$u_r(0) = 0, \quad p_r(0) = 0.$$

These are standard in axisymmetric problems and ensure both physical consistency and mathematical well-posedness of the PDE.

References

[1] Günther Grün and Martin Rumpf. Nonnegativity preserving convergent schemes for the thin film equation. *Numerische Mathematik*, 87:113–152, 2000.

[2] Xue-Li Wang, Jian Qin, and Peng Gao. Axisymmetric evolution of thin films in spin coating. *Journal of Fluid Mechanics*, 1007:A56, 3 2025.